

REVIEW

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Big data analytics in healthcare: data sources, tools, challenges, and opportunities

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Abstract

The increasing number of patients with chronic diseases and the concentration of medical resources have a substantial economic impact, leading to hospital visits, hospital readmissions, and additional healthcare expenses. Healthcare providers must now adopt big data strategies to keep up with the ever-increasing data deluge and enhance patient care. Therefore, this study aims to provide an overview of big data's role in healthcare prediction by discussing its data sources, analytical techniques, and challenges. Also, it proposes a conceptual framework to be adopted in real-time big healthcare data analytics. In addition, this paper answers the following questions. First, what are the most popular Big Data sources that can be utilized in healthcare prediction? Second, how can Big Data sources and Big Data processing frameworks be integrated to enhance healthcare prediction accuracy? The studies discussed in this paper have been presented in popular scientific research databases, such as IEEE, Springer, and Elsevier. Machine learning, deep learning, and healthcare are some terms used to search for these studies. Hence, a review of published papers utilizing machine learning and deep learning methods for the purposes of diagnosing, detecting, predicting, and monitoring conditions pertaining to healthcare has been conducted. The conducted review identified the key challenges, research directions, and recommendations for real-time healthcare prediction using machine learning technologies. In addition, the proposed conceptual framework can serve as a general methodology to be adopted by other researchers in the healthcare domain.

Keywords: Artificial intelligence, Big data, Deep learning, Healthcare prediction, Machine learning

Introduction

The healthcare industry is experiencing significant advancements in technology due to the rapid growth of technical and digital innovation. The magnitude of healthcare data is consistently growing and developing at an exponential rate. Given the available data, it becomes challenging to ascertain a suitable data type that can effectively provide valuable information. Consequently, the utilization of big data analytics (BDA) has been proven to be a valuable approach to the processing and visualization of data. The healthcare sector encompasses a wide range of sources that contribute to the presence of big data. Several examples of these sources are electronic healthcare data, registration data, medical imaging data, and management data. By

utilizing the provided data, valuable insights can be derived. BDA methodologies are utilized to analyze and integrate substantial volumes of unstructured, semi-structured, and structured data. This is done to offer various services to institutions, like hospitals and clinics, which are involved in critical decision-making processes [1].

The exponential growth of data is a problem for the majority of organizations, especially in healthcare. The large-scale data in the healthcare industry daily overwhelms a healthcare organization [2]. Healthcare can be improved by making better use of the huge amounts of patient-related clinical data that are now available [3]. Healthcare-related Big Data tasks involve datasets, physicians, and clinical. It would be too time-consuming or complicated to handle using conventional methods. The usage of sophisticated sensors in the healthcare industry is one example of this Big Data processing. Therefore, data science experts using Machine Learning (ML) algorithms assess and process Big Data [4].

Researchers have utilized Big Data to support their findings across various fields. BDA technology has been employed in the transportation sector to enhance service quality, customer satisfaction, and management processes [5]. The authors in [6] provided a comprehensive analysis of the utilization of BDA to enhance the strategic planning of airline routes. Big data has found application in the educational sector due to its potential to improve student motivation and performance [7].

A comprehensive review of the potential role of big data in healthcare prediction was introduced in this paper to answer the following questions: First. What are the most popular Big Data sources that can be utilized in healthcare prediction? Second. What BDA Tools can be used in healthcare? Finally, What are the challenges of using BDA in healthcare prediction?

The remainder of the paper is structured as follows: Sect. “[Background](#)” provides a background of big data including the history and their characteristics. Sect. “[Related Work](#)” goes through numerous sources of big data that can be used in healthcare forecasts. Sect. “[Sources of Big Healthcare Data](#)” explores Big Data Analytical Classification in Healthcare. Sect. “[Big Data Analytical Classification in Healthcare](#)” investigates the Big Data Processing architectures that can be used in healthcare. Sect. “[Big Data Processing Architectures](#)” describes the survey methodology and provides an overview of the literature on several Healthcare forecasting. Sect. “[Real-Time Big Healthcare Data Analytical Conceptual Framework](#)” proposes a conceptual framework for using Big Data in healthcare prediction. Sect. “[Big Data Challenges and Opportunities for Healthcare](#)” covers the challenges and prospects that face healthcare forecasting. Sect. “[Conclusion](#)” outlines and concludes the paper.

Background

Big Data is a general concept for an information processing architecture that spans many fields. The commercial, media, government, and healthcare sectors are among those gradually adopting Big Data into their information processing systems. Therefore, this section provides an overview of Big Data, and its characteristics, and explains how it can be adopted in healthcare information processing systems.

Characteristics of big data

A vast amount of data that has been continuously expanding for a very long period is referred to as "Big Data," and it is used to describe the development and usage of technologies that deliver the right data to the right person at the right time. In addition to coping with the complexity of managing more varied formats and connected data, the problem is dealing with the continuously expanding data volumes [8].

Big data describes the massive datasets that traditional computer programmers can't process. Big data can refer to structured, semi-structured, or unstructured data. Big Data encompasses not just massive amounts of data, but also its velocity, usefulness, heterogeneity, quality, and diversity. According to Laney in [9], volume, velocity, and variety are the three most frequently cited characteristics of Big Data. For quite some time, consensus has existed on the three Vs; nevertheless, some authors have extended them to include 10Vs [10], as shown in Fig. 1.

Big data in healthcare

Electronic health datasets that are too large, complex, and time-consuming to manage with conventional software and hardware, as well as traditional data tools and methods, are referred to as "Big Data" in the healthcare industry. The volume, variety, and administration requirements of Big Data in healthcare are all enormous [11].

The ability to store more data has forced nations and organizations to computerize medical records, leading to the creation of electronic health records (EHRs). The use of BDA in the field of healthcare facilitates the examination of large datasets culled from hundreds of patients, the identification of clusters and correlations between datasets, and the development of prediction models by means of ML. BDA has the potential to significantly improve the lifesaving services given by the healthcare sector. Tracking patient health, decreasing costs for healthcare organizations, identifying patients with chronic diseases by understanding the disease patterns of many patients, and avoiding human errors by using BDA can reduce errors, and prevent epidemics.

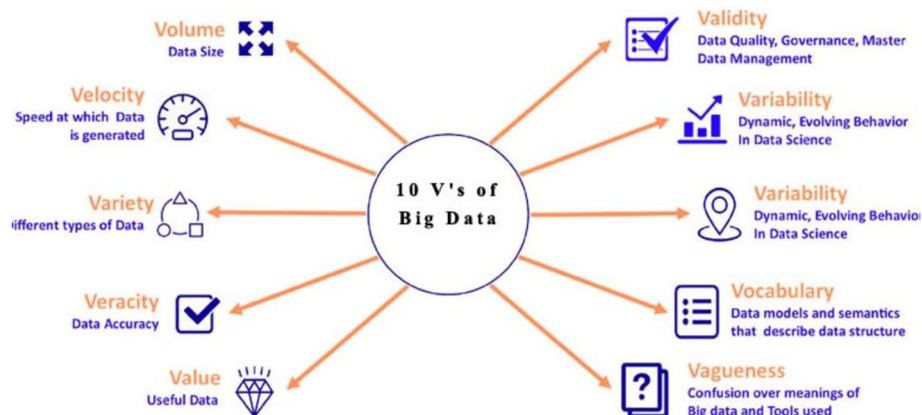


Fig. 1 10 V's of Big Data

Related work

Numerous research articles have demonstrated the benefits of BDA for healthcare companies and the types of data that can be studied with this approach. Fewer studies have indicated; however, how big data analysis technology is used in healthcare, or what the primary hurdles may be for an institution that is looking to incorporate big data into its system. This section summarizes several significant reviews in the field, discussing what each review adds to the conversation and what it focuses on. Finally, the related works can be summarized as shown in Table 1 which compares among related work in terms of their findings.

In [12], the authors provided a comprehensive overview of prior research on ML-based methodologies for predicting chronic diseases and certain instances of its implementations. The authors analyzed data from multiple studies along with the methodology and instruments utilized by the researchers. A wide variety of techniques for ML, such as support vector machines (SVM), random forests (RF), decision trees (DT), and naïve Bayes (NB), have been extensively employed. Additionally, various deep learning methodologies and hybrid models are used to develop chronic disease prediction models, offering efficient and reliable clinical decision-making frameworks. To improve the overall healthcare system, finally, they provided recommendations for boosting the predictive outcomes of chronic diseases.

A disease prediction system was developed by the authors of [13]. The diagnosis algorithm considers age, gender, and symptoms to diagnose a patient. Depending on severity, diet and exercise can reduce the disease's effects. ML classification algorithms do this. The authors use gradient-boosting classifiers for diabetes prediction and random forest classifiers for liver and heart disease prediction. Data entered by the user are immediately input into the disease model. Depending on the input, it returns to a binary value (0 or 1), indicating health or illness. The model result is then converted into a patient health statement and shown on the screen.

The authors of [14] reviewed the contributions of numerous authors to data collecting, analysis, processing, and visualization, highlighting the significance and potential of big data in industrial processing and healthcare applications. It delineates several opportunities and obstacles, including data cleansing, absent values, and outlier examination, alongside the uses and characteristics of big data. This thorough review additionally offered methods for dirty data detection and cleaning, as well as outlier detection, applicable across several domains. The data cleaning and outlier detection methods employ optimization techniques to address the optimal centroid selection issue and identify anomalous data.

In [15], the authors introduced a mobile application (CoFighter) designed for the prevention and management of COVID-19 and other associated pandemics in a multinational context. They leveraged the widespread presence of mobile smart devices in households to develop and deploy an Android application for COVID-19 and analogous pandemics. CoFighter offers users, governments, and healthcare professionals a platform to handle COVID-19 and future pandemics, while also assisting frontline health workers in effectively addressing these crises. The program operates on Android devices with version 4.2 or higher and is designed for managing the COVID-19 pandemic, economic crises, and analogous future pandemics.

Table 1 A comprehensive comparative study of the related work

Author	Year	Findings
Islam et al. [12]	2024	This study provided a comprehensive overview of prior research on ML-based methodologies for predicting chronic diseases by analyzing the data from multiple studies along with the methodology and instruments utilized by the researchers using SVM, RF, DT, and NB. Finally provided recommendations for boosting the predictive outcomes of chronic diseases
Reddy et al. [13]	2024	This study developed A disease prediction system, using gradient-boosting classifiers for diabetes prediction and random forest classifiers for liver and heart disease prediction
Rahul et al. [14]	2023	This study reviewed the contributions of numerous authors to data collecting, analysis, processing, and visualization, highlighting the significance and potential of big data in industrial processing and healthcare applications. The data cleaning and outlier detection methods employ optimization techniques to address the optimal centroid selection issue and identify anomalous data
Okereke et al. [15]	2023	In this study, the authors introduced a mobile application designed for the prevention and management of COVID-19 and other associated pandemics in a multinational context offers users, governments, and healthcare professionals a platform to handle COVID-19 and future pandemics, while also assisting frontline health workers in effectively addressing these crises
Srinivasan et al. [16]	2023	This study applied neural network models using UCI datasets to make heart disease predictions that achieved the best accuracy rate of 98.7% among the most up-to-date ways of forecasting cardiovascular problems
Ozcan et al. [17]	2023	In this study, the authors applied the classification and regression tree (CART) algorithm using Cleveland, Hungarian, Switzerland, Long Beach VA, and Statlog dataset to predict heart disease. The model accuracy 87% prediction
Pan, et al. [18]	2022	This study employed a different ML algorithm with UCI dataset. It uses ten various performance measures that show that the categorical features work better than the numerical features and the combined features
Pathan, et al. [19]	2022	This study analyzed how different feature selection methods affect the efficiency of ML models. Using Framingham dataset, authors utilized a filter-based feature selection method, specifically the ANOVA-F test, to extract the most relevant features from the datasets. Authors conclude that feature selection improves prediction accuracy
Bataineh, et al. [20]	2022	In this study, authors developed an alternate training technique for (MLP) that employs a (PSO) algorithm for the identification of heart disease. The suggested MLP-PSO surpasses all previous algorithms, with an accuracy of 84.61%
Rehman et al. [34]	2022	Developed a ML model through the compilation of a DNA microarray dataset, to discover discriminative features that influence the categorization of diverse cancer types and aid in early diagnosis
Yousef et al. [24]	2021	Analyzed different types of technologies, tools, and applications that are utilized for data integration in the healthcare industry with discuss the challenges that faced when integrating massive volumes of healthcare data
Bakker y et al. [33]	2020	Utilized two deep learning-based survival prediction models were put forth to help doctors choose the best breast cancer treatments and steer clear of inefficient ones
Uddin et al. [27]	2020	The authors conducted a study presents an overview of the current significance of big data in the COVID-19 era and how these applications correspond with important BDA models
Greasley et al. [32]	2019	Proposed prediction models predicated on a single diagnostic feature could make it possible to diagnose brain illnesses prior to the manifestation of noticeable symptoms
Benhlma et al. [23]	2018	Introduced an evolution of healthcare big data and the data mining methods employed are analyzed, along with their general and healthcare-specific applications
Sterling h, et al. [21]	2017	Presented the results of a survey that delves into the application of big data in healthcare and the possibilities it offers
Grover et al. [22]	2016	Compiled 205 publications from reputable journals that have presented and addressed BDA in the healthcare industry
Zhang et al. [28]	2016	In this study, the authors provided a thorough analysis of what has been accomplished so far and ideas for how BDA
Bachiri et al. [29]	2016	The authors discussed how big data can be used in the areas of pharmacy, pharmacology, and toxicology. It looks at how experts have used BD to solve problems and find answers

Table 1 (continued)

Author	Year	Findings
Guyon et al. [30]	2016	The authors proposed a framework that employed an SVM classifier to conduct parameter tuning. The main objective of this framework was to forecast diseases. The accuracy of the approach's implementation was 90.6%
Wang et al. [45]	2015	The authors examined many effective predictive analytics methods for cardiovascular disease, hypertension, and diabetes. The authors also claimed that real-time data analytics in healthcare will allow hospitals to detect infections before they spread
Evans et al. [25]	2015	The authors suggested the Smart Health concept as a method to connect all platforms and create a uniform standard learning healthcare system
Calvert et al. [26]	2015	The authors examined the application of ML, big data, and blockchain technology in the fields of medicine, healthcare, public health surveillance, and case prediction
Pelegris et al. [31]	2010	Proposed a Big Data system, utilizing the Hadoop framework, was put up with the aim of identifying early indications of illness
Shvachko et al. [35]	2010	The authors investigated the use of big data in the field of digital health and explored the nature of health data, The authors acknowledged the necessity of proposing a methodology for developing big data analysis in the health field

In [16], the authors used neural network models like Naïve Bayes and Radial Basis Functions using the UCI dataset to make heart disease predictions that were 94.78% and 90.78% accurate, respectively. Learning vector quantization had the best accuracy rate of 98.7% among the most up-to-date ways of forecasting cardiovascular problems. Early intervention, personalized medicine, technology advances, the effect on public health, and ongoing research are some of the most important contributions. Together, they make CHD easier to deal with for both individuals and society as a whole.

In [17], the authors utilized the classification and regression tree (CART) algorithm, a supervised ML method using Cleveland, Hungarian, Switzerland, Long Beach VA, and Statlog dataset to predict heart disease and identify decision rules by analyzing correlations between input and output variables. The study identifies the most important factors impacting heart disease. The model's reliability is validated by its 87% prediction accuracy across all performance criteria. The study's abstracted decision principles simplify clinical application and require no further knowledge. The suggested approach can benefit both healthcare professionals and individuals facing cost and time restrictions while diagnosing and treating cardiac disease.

The authors of [18] employed a different algorithm, including Gradient Boosting, Extreme Gradient Boosting (XGBoost), AdaBoost, and CatBoost. It also looked at artificial neural networks, random forest, support vector machines (SVM), decision trees, and logistic regression. To forecast heart disease, a soft vote ensemble system with learning algorithms has also been put in place. The UCI dataset was used as a standard in this study. It is free to use and is widely used. It uses ten various performance measures that show that the categorical features work better than the numerical features and the combined features. It was also found that the best results for predicting heart disease come from using SVM and AdaBoost classifiers together, which use categorical features.

In [19], the authors analyzed how different feature selection methods affect the efficiency of ML models. The publicly accessible datasets for cardiovascular disease and Framingham heart disease were the subjects of this investigation. An initial data pre-processing stage was carried out in this study, which included data cleansing, balancing, and transformation. Secondly, to effectively forecast cardiac illness, the authors utilized

a filter-based feature selection method, specifically the ANOVA-F test, to extract the most relevant features from the datasets. The impact of features on the accuracy of predictions made by different ML prediction models was also investigated in the classification tests, which were run with both the complete and reduced feature sets. Authors conclude that feature selection improves prediction accuracy while reducing computing complexity by selecting only the most significant features associated with heart disease.

In [20], the authors developed an alternate training technique for multilayer perceptron (MLP) that employs a particle swarm optimization (PSO) algorithm for the identification of heart disease. This study employs the suggested MLP-PSO hybrid algorithm alongside ten distinct ML methods to detect the disease. The suggested MLP-PSO surpasses all previous algorithms, with an accuracy of 84.61%. The findings indicate that the new MLP-PSO classifier allows practitioners to identify heart disease with greater timeliness, precision, and efficacy.

The authors of [21] presented the results of a survey that delves into the application of big data in healthcare and the possibilities it offers. It looks into the different types of data that can be collected and used for analysis. The many challenges posed by large healthcare datasets are also addressed.

In [22], the authors compiled 205 publications from reputable journals that have presented and addressed BDA in the healthcare industry. Protecting healthcare patients' data in the era of big data is a pressing concern.

In [23], the authors provided a comprehensive overview of the tools and methods employed in the development of analytical applications for healthcare. The evolution of healthcare big data and the data mining methods employed are analyzed, along with their general and healthcare-specific applications. Health analytics solution essentials such as platforms and technologies are also discussed.

The authors of [24] analyzed many different types of technologies, tools, and applications that are utilized for data integration in the healthcare industry. In addition to this, they discuss the challenges that are now faced when integrating massive volumes of healthcare data and investigate the possibility of future research opportunities in this sector.

The authors of [25] introduced a review of BDA systems utilized in healthcare, highlighting the numerous algorithms, strategies, and tools that can be deployed in cloud, wireless, and Internet of Things contexts. The authors suggested the smart health concept as a method to connect all these platforms and create a uniform standard learning healthcare system.

Within the context of the COVID-19 pandemic and other epidemics, the authors of [26] examined the application of ML, big data, and blockchain technology in the fields of medicine, healthcare, public health surveillance, and case prediction. In addition to this, it discusses potential obstacles that may be faced by medical experts and health technologists while developing forward-looking models to improve the quality of human life.

In [27], the authors conducted a literature review on the uses of big data in the context of the COVID-19 pandemic. The authors focused primarily on the use of big data applications in four critical areas, one of which was healthcare. The study presents an overview of the current significance of big data in the COVID-19 era and how these applications correspond with important BDA models by comparing the use of big

data apps before and during the pandemic. This comparison is made by contrasting the use of big data applications before and during the pandemic.

The authors of [28] provided a thorough analysis of what has been accomplished so far and ideas for how BDA might be used in healthcare organization management to better meet the growing demand for medical care.

In [29], the authors looked at how big data can be used in the areas of pharmacy, pharmacology, and toxicology. It looks at how experts have used BD to solve problems and find answers. Comparative analysis is used in the study to look at how big data is used in these three areas.

The authors of [30] employed an approach for health prediction that involved the utilization of the EHR platform and investigation methodologies for consumer social health profile information. a proposed framework that employed a Support Vector Machine (SVM) classifier for the purpose of conducting parameter tuning. This was done to enhance the accuracy of classification and increase the efficiency of linking Spark and Microsoft Azure-based streaming data from wearable devices. The main objective of this framework was to forecast diseases. The accuracy of the approach's implementation was 90.6%. The authors emphasized the need to employ deep learning techniques, namely within a cloud-based framework that can effectively handle a comprehensive dataset consisting of disease records.

The authors of [31] proposed a Big Data system, utilizing the Hadoop framework, was put up with the aim of identifying early indications of illness. This methodology combines data obtained from neuropsychological assessments with noninvasive techniques such as magnetic resonance imaging (MRI) and magnetic resonance spectroscopy (MRS).

In [32], the authors proposed that the development of prediction models predicated on a single diagnostic feature could make it possible to diagnose brain illnesses prior to the manifestation of noticeable symptoms.

The authors of [33] utilized two deep learning-based survival prediction models were put forth to help doctors choose the best breast cancer treatments and steer clear of inefficient ones.

In [34], the authors developed a ML model through the compilation of a DNA microarray dataset, with the goal of discovering discriminative features that influence the categorization of diverse cancer types and aid in early diagnosis.

The authors of [35] investigated the use of big data in the field of digital health and explored the nature of health data, its particular attributes, and the technology accessible for its use. The authors acknowledged the necessity of proposing a methodology for developing big data analysis in the health field. They also emphasized the need to construct a new, adaptable architecture that can effectively address the problems discussed in their review.

Disease prediction is a major healthcare sector concern, according to [36]. Experts use various data mining approaches to diagnose diseases due to the rising global death rate. Thus, the authors examined many effective predictive analytics methods for cardiovascular disease, hypertension, and diabetes. The authors also claimed that real-time data analytics in healthcare will allow hospitals to detect infections before

they spread, closely track therapy progress, and make more informed treatment decisions to reduce illness and death.

Sources of big healthcare data

Due to the expansion of information in the medical industry, there has been a huge increase in electronic data in the fields of medicine and health [37]. The large set of these data consists of data about disease diagnosis and therapy for the patients, medicine prescription records, information from clinical medical tests, and information on physical health [38]. Several different sorts of data are produced when a patient's health condition is continuously monitored. It is possible for medical data to be only somewhat organized, like the logs that certain medical devices generate. Either structured, as in the case of the industry standard EHRs, or unstructured, as in the case of the data generated by biological images [39].

Electronic health records

EHRs are electronic copies or digital formats that store a complete patient's medical history. EHRs may include data on the outcomes of administrative and clinical consultations between the patient and the doctor. Additionally, patient-centered records that include real-time data contain data that is instantaneously accessible [40]. EHRs can contain a range of information, including current medical conditions, demographics, medical history, prescriptions, results of laboratory tests, radiographic images, and individual details like age and weight [41].

Patient-reported outcomes (PROs)

Defined as an individual's report on his or her health and treatment decisions based on that person's level of medical expertise, symptoms, health status, and quality of life in relation to health are a variety of outcomes included in this account or report [42].

Data collected from wearable sensors

Biochemical, physiological, and motion-sensing data can all be gathered with the help of wearable sensors that track a variety of health markers, including body temperature, blood pressure, and heart rate. These sensors are crucial for keeping track of a patient's health. As a result, it has the capacity to gather and share data about patient health. This method of data analysis, when combined with EHRs, can help in the early diagnosis, and continued monitoring of chronic diseases [43].

Data extraction from social networking

For carrying out a social network analysis (which includes Twitter, Facebook, Instagram, and other similar platforms), data are collected from social media platforms, such as social networking sites. It is possible to collect information regarding disease patterns, patient satisfaction, pleasure, interests, and moods by using the posts that patients make on social media platforms that are accessible online [44].

Data collected from mobile phones

Mobile phones have developed from basic communication tools to complicated devices with a wide range of functions and services. Cameras, accelerometers, and GPS systems are just some of the sensors currently available. Mobile phones are convenient for collecting health information because of their widespread use and extensive range of features. They have allowed the growth of several useful healthcare applications, such as those for tracking pregnancy [45], providing feeding to infants [46], and checking heart rates [47].

Big data analytical classification in healthcare

Due to their dynamic nature and complexity, several types of healthcare data, including medical pictures, biological signals, written prescriptions, and structured data from EHRs—can be challenging to interpret. The term "Big Data analytics" refers to the practice of using technological tools to collect a large amount of diverse data of high quality for further modeling, and analysis. The healthcare industry employs many forms of BDA, as depicted in Fig. 2, covering descriptive, predictive, and prescriptive analytics. This section delves into the particulars of each analysis type and how it is utilized within the healthcare domain.

Descriptive analytics

Collects historical data to respond to questions, finds trends in current operations, and prepares this data for future research. A straightforward "What happened?" question served as a succinct summary. Population health management relies on analytics like these to determine the total number of patients in a given area or pinpoint places where services could be enhanced [48].

Predictive analytics

Used to forecast and comprehend potential future events. It examines previous data patterns and trends to forecast likely future occurrences. It might be summed up by asking "What might happen in the future?". This type of analytics can be used to forecast the weather, identify populations or groups in need of healthcare, anticipate load failure, and forecast epidemics and breakouts [49].



Fig. 2 Classification of Big Data analytics in healthcare

Prescriptive analytics

It is an innovative method of analyzing data that helps decide the best course of action by combining the results of descriptive and predictive studies. Beyond just forecasting the future, it also helps in choosing the best options for preventive measures to reduce risks. The question "What should we do?" is answered by the predictive analysis [48].

Big data processing architectures

Big Data architectures have been defined in a variety of ways by researchers who focus on BDA. Streaming Graph, Hadoop, and MapReduce, among others, are examples of significant architectures [48]. The most well-liked solutions for Big Data healthcare analytics are shown in Fig. 3 and listed below.

Batch processing

Batch processing manages massive amounts of data through parallel computations executed according to a determined schedule which can cause latency ranging from minutes to hours.

Apache hadoop

One of the significant frameworks inside the Apache platform is Hadoop, which was developed by Doug Cutting and Apache Lucene. The aforementioned is a compilation of open-source software tools that are employed for the purpose of distributed computation, processing, and storing of extensive datasets, commonly referred to as big data. Hadoop encompasses two fundamental architectural components, namely, the Hadoop Distributed File System (HDFS) and the MapReduce framework [50]. HDFS is a

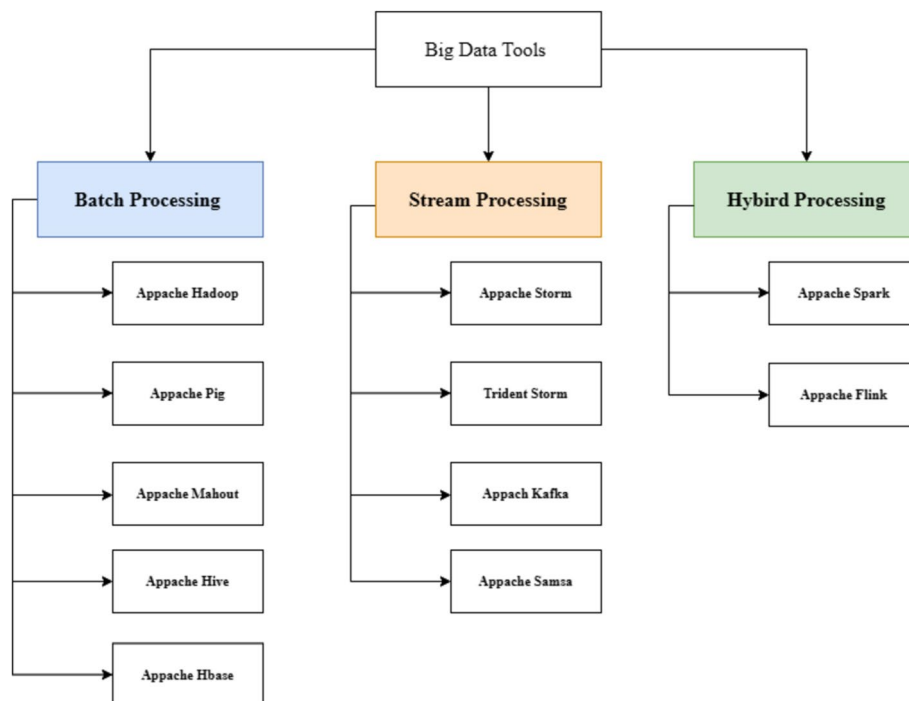


Fig. 3 Classification of Big Data analytics in healthcare

Java-based Apache distributed open-source processing platform for Big Data technologies that employs primary data storage to handle data processing and proposal storage for Big Data applications. By dividing enormous amounts of data into smaller chunks and distributing them across numerous servers and nodes, it improves the healthcare data analytics system and provides the Hadoop cluster with the necessary underlying storage. Name-node and Data-node make up HDFS's architecture, with Name-node serving as the master and Data-node as the slave. Documents with a 64 M size limit are kept in a data node that cannot be modified [50]. Hadoop's MapReduce architecture is a popular functional programming processing and analysis system. It helps divide huge jobs, collect results, and analyze massive databases in parallel. Data analysis and processing using map and reduce phases. MapReduce's main components are the Client, Job-Tracker (Master Node), and Task-Tracker (Slaves Node). Jobs are submitted to the job-tracker as JAR files. Job-Tracker tracks all MapReduce jobs as the master service. Job-Tracker provides duties to Task-Tracker, which performs them like a slave. Real-time, graph, and interactive processing are impossible with MapReduce [51].

Apache pig

Pig is a data flow system designed by Yahoo at a high level, mostly utilized for composing uncomplicated queries. These searches are subsequently transformed into MapReduce programs and performed across a Hadoop cluster. Pig has been utilized as a solution to address the complexity associated with Map and Reduce Stages. By reducing the amount of time required to write MapReduce applications, Pig facilitates the processing of extensive datasets efficiently [52].

Apache mahout

ML library Apache Mahout is free and open source. Mahout executes MapReduce algorithms on Hadoop servers and other platforms. Mahout is a set of Java libraries with several powerful analytical tools. Categorization, classification, text mining, and collaborative filtering libraries are available. K-means, fuzzy K-means, and mean shift libraries (for text scanning and contextual data assignment) enable clustering [53].

Apache hive

Hive analyses and queries enormous Apache Hadoop data warehouse collections. HiveQL manages this huge data set. To simplify data querying, Hive's Extract Transform Load writes data using Structured Query Language. The most common uses of Hive include data encapsulation, analysis, ad hoc querying, and summarizing large data collections. Hive has four main components: Clients, Services, Processing Framework, and Distributed Storage. Thrift, JDBC, and ODBC clients can be written in any Hive-supporting language, such as C++, Java, Python, etc. Services are for questions. Hive services include a command line interface, Web interface, server, driver, meta-store, and more. Our Hadoop MapReduce handles query processing, execution, and maintenance. All network node data is permanently preserved in HDFS [54].

Apache Hbase

The "no-relational database" HBase can store structured Big Data for big tables. Data are organized in rows and columns in tables. These tables support millions of rows and thousands of columns. Column families organize several HBase attributes and keep their components together. Apache HBase's key components are the HMaster Server, Region Server, and Zookeeper. HMaster Server does database activities including putting new rows into a table and maintaining HBase Region Servers. Hbase Region Servers manage HBase table regions. Zookeeper manages HBase [55].

Stream processing

Stream processing works with data that is updated in real-time and needs to be processed immediately upon arrival, which can cause latency ranging from milliseconds to seconds.

Apache storm

Open-source real-time data analysis framework. Bolts and Spouts Bolt produces output streams from input streams, while Spouts can produce or retrieve data from data queues. A storm program's structure resembles a bolt spout. The nimbus's master, worker, and zookeeper nodes make up Storm. Master nodes oversee and distribute topology execution, while worker nodes execute spouts and bolts. Finally, the zookeeper synchronizes scattered coordination [41].

Trident storm

The utilization of the Trident APIs is a prominent feature of the Apache Storm topology. To enhance productivity, Trident APIs employ a strategy of decomposing tasks into smaller clusters. The optimization of performance and reduction of delay can be achieved by the adjustment of the batch size. Nevertheless, it is worth noting that directed acyclic graphs (DAGs) are not suitable for executing iterative algorithms due to their inherent topological characteristics [56].

Apache Kafka

Kafka is an open-source distributed platform that facilitates communication through a publish-subscribe messaging system. The first development of this technology took place at LinkedIn, subsequently leading to its integration into the Apache project. As a result of its notable attributes like scalability, durability, and fault tolerance, numerous enterprises have adopted its usage [57].

Apache Samza

Apache Samza is a distributed stream processing framework that is designed to handle large-scale data processing tasks. Apache Samza is a software framework that was developed by LinkedIn and is constructed upon the foundations of two existing frameworks, namely, Kafka and Hadoop. Apache Samza was created with the purpose

of addressing many challenges in stream processing, such as scalability and resource allocation [58].

Hybrid processing

Often both, minimal processing time and high accuracy are required. Because batch and stream processing are unable to accomplish this, hybrid processing integrates the two methods.

Apache spark

When compared to numerous other tools like Hadoop and Storm, Spark is thought to be an efficient framework. Among the data stores, it can access are HDFS and HBase. Attractive characteristics of Spark include the Mllib library's efficient and quick iterative ML methods, Hive's structured data analysis, GraphX and SparkSQL's graph processing, and Spark's ability to restore data from different sources and change it using SQL languages. Spark divides data streams into digestible chunks prior to transforming them into a collection of resilient distributed datasets (RDDs) referring to as Discretized Stream (Dstream) [59].

Apache Flink

Apache Flink is an open-source framework that facilitates the execution of iterative algorithms, batch processing, and real-time data-stream pipelines at no cost. One of the main benefits of employing a distributed system lies in its ability to efficiently handle extensive data sets while minimizing the delay and exhibiting a robust fault tolerance. The utilization of the Dataset API is necessary when processing data in batches that are divided by a specific threshold [60].

A significant portion of BDA research shows the efficiency of BDA tools in monitoring specified antenatal data for babies small for gestational age, assessing the response to COVID-19 in Taiwan, and applying BDA in mental health care [61–63]. Islam et al. [64] introduced a case study by combining a database, a mobile app, a web app, and the creation of a new platform for patients and doctors to communicate with each other. Furthermore, the platform aids in storing patients' health data to make the final forecast utilizing ML methods, allowing for the correct healthcare treatment to be provided by both humans and computers. This platform achieved 95% accuracy, and it can do it at a cost that is 90% less than the local hospital. ISMAIL et al. [65] utilized a Cluster Analysis (CA) model, which statistical method used in Big Data to attempt to find hidden patterns and "clusters" in large data sets. By utilizing healthcare applications to discover patterns of cost change among end-stage renal disease patients who started hemodialysis using several clustering methods. Nisar et al. [66] evaluated hospital performance using a healthcare application across a range of quality metrics, analytical tools called graph analytics are used to determine the direction and strength of relationships between elements in a graph. Understanding, encoding, and visualizing pairwise associations between two objects at a time as well as the structural characteristics of the graph overall are the main goals of graph analytics in big data.

Real-time big healthcare data analytical conceptual framework

In this section, a real-time big healthcare analytical conceptual framework is proposed. It is important to note that the proposed architecture is flexible enough to deal with a wide range of use cases, from emergency medical care to early disease identification. Also, it was designed to handle massive medical data from various sources and various formats in real time by using artificial intelligence techniques to predict diseases in the early stages. As depicted in Fig. 4, the suggested framework consists of five primary phases: data collection, data storage, data transformation, streaming pipeline procedure, and presentation.

Data collection phase

The first and most essential phase of Big Data applications in healthcare prediction is the collection of data from diverse types, structured data, such as EHRs and patient records; semi-structured data, such as the logs of health sensors devices; and unstructured data, such as clinical photographs, are all obtained at this stage. Clinical data are collected from various internal and external sources and swiftly inserted into the appropriate databases, based on their content type.

Data storage phase

In the data storage phase, the collected data is saved and ready for subsequent phases. The utilization of traditional data storage techniques may not be appropriate for managing systems that include sensitive information, due to the possibility of its inclusion within the gathered data. Therefore, it is imperative to employ Big Data storage

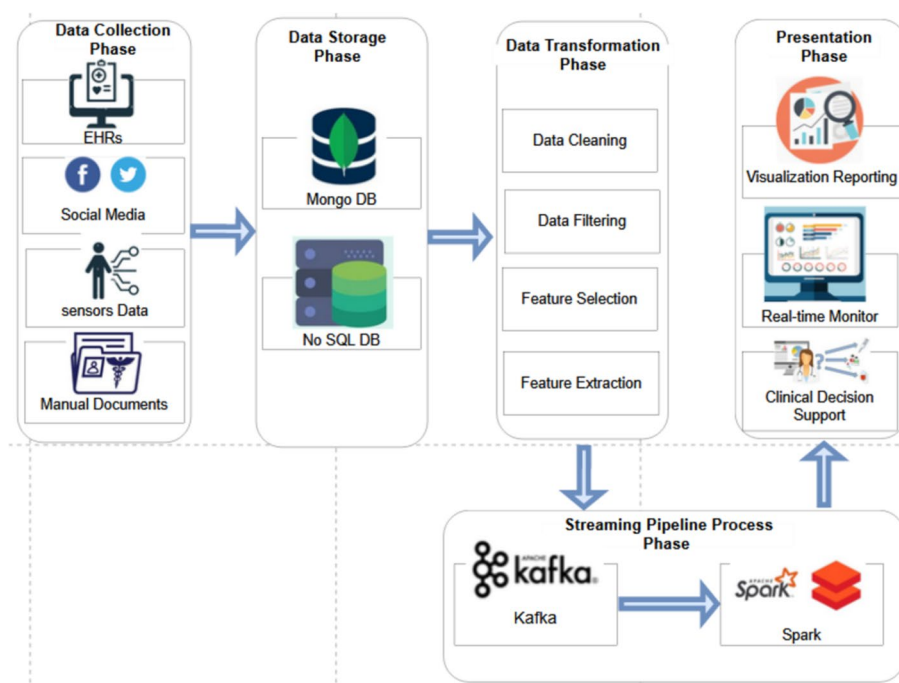


Fig. 4 Real-Time Big Healthcare Data Analytical Conceptual Framework

frameworks, such as NoSQL databases or MongoDB, during this phase. As a result, it demonstrates improved scalability and offers a greater storage capacity.

Data transformation phase

The data transformation phase is vital because it prepares the data, hence reducing the need for costly additional resources when used for Big Data. To generate precise predictive models and increase the dependability of artificial intelligence, it is advised to ensure that the data is prepared properly. This phase consists of four steps which can be summarized as follows: Data Cleaning is the first step which includes several elements such as managing missing data, noise reduction, and normalization. The second step is Data Filtering which involves omitting material that is not pertinent to healthcare monitoring according to some criteria. The next step is Feature Extraction, which entails extracting fewer attributes than the initial set. Finally, the chosen features will be incorporated into the a priori model.

Streaming pipeline process phase

During this phase, the data obtained from various sensors can be integrated with the trained and tested models to enhance the management, monitoring, and prediction of data flow. The utilization of BDA frameworks helps accelerate the generation of real-time predictive information. Apache Spark, a highly potent BDA technology, can be effectively employed at this phase, as previously mentioned in this document. Apache Spark incorporates a variety of high-level libraries for ML, stream processing, and structured data processing. These libraries enhance the intelligence and effectiveness of healthcare prediction.

Presentation phase

The presentation phase yields a diverse range of outputs, including reports pertaining to prediction decisions and the display of data related to patient health monitoring. Within the realm of healthcare, the visualization of real-time data and its prioritization play a pivotal role in enhancing decision-making processes and mitigating potential catastrophes. Significant data encompasses patient dashboards used to monitor daily health status, instantaneous alarms, and proactive messaging generated through projections.

Big data challenges and opportunities for healthcare

Nowadays, the healthcare industry generates a large quantity of data from a variety of sources, including quantitative sources such as sensor data and gene arrays and qualitative sources such as demographics and free text [67]. Medical data is sensitive and more difficult to collect than other types of data [68]. Therefore, this section will address the several obstacles encountered by big healthcare data analytics approaches in the healthcare industry and propose potential solutions for these issues.

Real-time processing issues

Using unmanaged data sources, such as social networks, may probably have a negative effect on data. The validity of the analysis may also be significantly impacted by any coding errors. Therefore, the consistency, quality, and dependability of the data present

additional challenges for big data. Furthermore, real-time data streaming techniques are becoming increasingly important since clinical data is generated continuously [11]. Since most individuals use social networks, a massive amount of data will be produced quickly and in a variety of ways. Utilizing and integrating this data to enhance healthcare outcomes is therefore both very interesting and difficult. Hence, developing a medical decision-making system to assess data from social media sites like Facebook and Twitter to predict the development of diseases in various geographic places is one potential solution.

Data ownership

Although medical records are owned by healthcare institutions, the information contained in them is not. One major issue with the use of Big Data applications in healthcare and other sectors is who owns the data [69]. Patients, on the other side, consider themselves to be the owners of this information. This problem can be solved by requiring healthcare providers to obtain written permission from patients before using their data for research purposes.

Privacy and security

There may be sensitive or private information in the medical records and data obtained from patients [70]. The majority of BDA and data processing technologies are open source and have no built-in safeguards [71]. Encryption and other security techniques are used to overcome these challenges and maintain privacy.

Storage and processing issues

The primary obstacle linked to Big Data is clearly the storage and analysis of an extensive volume of information. Whenever a new storage technology is released, the amount of data generated by social media platforms, healthcare providers, and financial markets increases significantly [72]. Furthermore, these datasets are not only extensive in size but also comprise often generated data of many categories. The primary challenge in managing Big Data is the inadequate capacity of current systems to store and process data of such magnitude and complexity [73]. Hence, cloud computing can be employed to tackle storage concerns. This solution can be utilized by small and medium-sized hospitals and care groups to effectively address their financial and data storage challenges.

Conclusion

Big Data refers to vast datasets that are beyond the processing capabilities offered by conventional processing techniques. To enhance effective decision-making, institutions are increasingly adopting "Big Data analytics", which entails analyzing a vast volume of data to discover previously unidentified patterns. This study performed a comprehensive analysis of Big Data in healthcare, including its defining features, data sources, analytical approaches, and platforms employed. This paper also proposed the adoption of a conceptual framework for real-time big healthcare predictive analytics. Furthermore, it examined the challenges that big data encounters in the healthcare sector and provides insights into the potential effective utilization of big data in healthcare. This study can serve as a blueprint for other researchers in the healthcare domain who are interested

in exploring new search patterns and addressing hurdles related to BDA in their own investigations.

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MB wrote the main text of the manuscript; NR and HAH revised the manuscript. All authors read and approved the final manuscript.

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Competing interests

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